

Route G-R7: One Clock Action, Frequency, Phase and Path Correlations

A conservative quantum extension of the G-R6 baseline

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Abstract

Preserve the E2/E3 energy gaps, thermal response and metric of G-R4–G-R6. Promote the path label to a coherent two-branch degree of freedom. The same proper-time action gives frequency, relative internal phase, and ideal interferometric visibility. An explicit pure/mixed counterexample shows why these three observations alone need not certify entanglement. Joint observables distinguish the cases. The full scalar interferometer phase, radiative environment and apparatus closure remain independent missing inputs; no time law, experimental signal or hardware is invented.

1 One conservative effective action

The retained local Hamiltonian, in the basis (g, e_2, e_3) , is

$$H_C(T) = \sum_{i=2,3} h\nu_i^0 [1 + b_i(T)] |e_i\rangle\langle e_i|, \quad b_i(T) = \beta_i u(T), \quad u(T) = a_{\text{rad}} T^4. \quad (1)$$

Here $\beta_i = -\Delta\alpha_i/(2h\epsilon_0\nu_i^0)$ and all pinned G-R4 inputs are unchanged. The static scalar dipole approximation is retained, not promoted to a complete radiative model. For a prescribed branch $r = A, B$, use the effective action

$$S_r = S_{\text{guide},r} + \int_r \left[i\hbar\chi^\dagger \partial_\tau \chi - E_0(x) - \chi^\dagger H_C(T(x))\chi \right] d\tau. \quad (2)$$

E_0 includes ground-state rest energy and any physical common ground-level shift. G-R4's differential polarizabilities do not determine that ground shift separately. Guide and pulse phases are explicit inputs, not derived from the thermal gap data. We assume state-independent guides and coincident final wave packets at the retained order; otherwise the branch evolution must include the external motion as well.

Variation of the internal state gives $i\hbar\partial_\tau\chi = H_C\chi$. For diagonal adiabatic evolution during the analyzed hold,

$$U_r = \text{diag}(1, e^{-i\theta_{2r}}, e^{-i\theta_{3r}}), \quad \theta_{ir} = 2\pi\nu_i^0 \int_r [1 + b_i(T_r)] d\tau_r. \quad (3)$$

Writing $\lambda_r = d\tau_r/dt$, the received frequency against reference R is $\nu_{i,r}^{(R)} = (\lambda_r/\lambda_R)\nu_i^0(1 + b_i)$, recovering G-R6. Thus frequency is the phase derivative, not a second adjustable law. General paths require all segments, kinematic terms, preparation phases and pulses. The fixed reference is not a third entangled quantum system here. This extension uses the established clock-interferometry framework [1, 2], not a new quantization rule.

2 Joint evolution fixes phase and visibility

For a balanced path state and initial internal state ρ_C , the joint evolution is controlled by the path. Define

$$\mathcal{U} = |A\rangle\langle A| \otimes e^{i\zeta_A} U_A + |B\rangle\langle B| \otimes e^{i\zeta_B} U_B, \quad \zeta = \zeta_A - \zeta_B. \quad (4)$$

ζ_r includes the scalar propagation, ground-energy, guide and beam-splitter contributions. Tracing the clock gives

$$\rho_P = \frac{1}{2} \begin{pmatrix} 1 & e^{i\zeta} \kappa \\ e^{-i\zeta} \kappa^* & 1 \end{pmatrix}, \quad \kappa = \text{tr}(\rho_C U_B^\dagger U_A). \quad (5)$$

Projection onto $(|A\rangle + e^{i\varphi}|B\rangle)/\sqrt{2}$ yields

$$P_+(\varphi) = \frac{1}{2} [1 + \Re(e^{i(\varphi+\zeta)} \kappa)], \quad \mathcal{V} = |\kappa|, \quad \phi_{\text{fringe}} = \zeta + \arg \kappa. \quad (6)$$

The fringe phase is undefined at $\kappa = 0$. Unequal path populations $p_A, 1 - p_A$ instead give $\mathcal{V} = 2\sqrt{p_A(1-p_A)}|\kappa|$. Only the internal contribution $\arg \kappa$ is fixed without ζ . Moving a scalar term $s_r(\tau)I$ into H_C and out of E_0 changes U_r and ζ_r oppositely; the product in Eq. (6) is invariant. One cannot simply discard physical branch-dependent ground energies as an unobservable overall phase.

Separate two-level preparations. Let $\delta_i = \theta_{iA} - \theta_{iB}$ and $|\chi_0\rangle = \sqrt{1-p}|g\rangle + \sqrt{p}e^{i\eta}|e_i\rangle$. Then

$$\kappa_i = (1-p) + pe^{-i\delta_i}, \quad (7)$$

$$|\kappa_i|^2 = 1 - 4p(1-p)\sin^2(\delta_i/2). \quad (8)$$

For $p = 1/2$ this becomes

$$\kappa_i = e^{-i\delta_i/2} \cos(\delta_i/2), \quad \mathcal{V}_i = |\cos(\delta_i/2)|, \quad 1 - \mathcal{V}_i = \delta_i^2/8 + O(\delta_i^4). \quad (9)$$

An energy eigenstate has visibility one: a different scalar phase is not itself a which-path record in the internal state. For a pure initial clock the joint state has Schmidt eigenvalues $(1 \pm |\kappa|)/2$, hence negativity

$$\mathcal{N} = \frac{\sqrt{1-|\kappa|^2}}{2}. \quad (10)$$

This inference requires purity and the declared closed unitary model.

One shared ground is not two clock qubits. For a coherent equal three-level preparation,

$$\kappa_{023} = \frac{1 + e^{-i\delta_2} + e^{-i\delta_3}}{3}, \quad (11)$$

not $\kappa_2\kappa_3$. E2/E3 runs or a genuine qutrit preparation must be specified. No independent tensor factor is invented for each gap.

3 A counterexample and an actual entanglement discriminator

Since $U_B^\dagger U_A$ is diagonal in this model,

$$\kappa = \sum_j (\rho_C)_{jj} e^{-i\delta_j}, \quad \delta_g = 0. \quad (12)$$

Erasing initial energy coherences leaves the entire reduced path signal unchanged, but can remove joint entanglement. A sharp two-level fixture is $U_A = I$, $U_B = Z$, with

$$|\Psi_{\text{pure}}\rangle = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle - |11\rangle), \quad (13)$$

$$\rho_{\text{mixed}} = \frac{1}{2}|+\rangle\langle+| \otimes |0\rangle\langle 0| + \frac{1}{2}|-\rangle\langle-| \otimes |1\rangle\langle 1|. \quad (14)$$

The inputs are respectively $|+\rangle$ and $I/2$ for the clock. Both output path states equal $I/2$, so both have zero visibility. The pure state has negativity $1/2$; the explicitly separable mixture has negativity zero. Frequencies and the diagonal conditional phase map are the same. Visibility alone therefore does not certify entanglement.

One joint witness for this calibrated fixture is

$$W = \frac{1}{2}(I - X_P \otimes Z_C - Z_P \otimes X_C). \quad (15)$$

For product Bloch vectors a, b , $a_x b_z + a_z b_x \leq \sqrt{a_x^2 + a_z^2} \sqrt{b_z^2 + b_x^2} \leq 1$; convexity extends $\langle W \rangle \geq 0$ to separable states. The pure fixture gives $-1/2$ and the mixed fixture gives zero. This is a joint-measurement specification, not a realized π -phase experiment or a witness automatically optimal for a small phase.

State-resolved fringes and their limitation. With populations p_j unchanged in both arms, joint port/energy probabilities are

$$P(+, j) = \frac{p_j}{2}[1 + \cos(\varphi + \zeta - \delta_j)]. \quad (16)$$

Differences between excited and ground fringe phases isolate $-\delta_i$ if external phases are state-independent and pulses calibrated. This is linear phase information, but still does not distinguish the pure/mixed counterexample. Clock-coherence measurements and joint noncommuting observables, or tomography, are needed for an entanglement claim.

Reversibility is not an arbitrary contrast correction. A common final clock unitary V preserves κ because $(VU_B)^\dagger VU_A = U_B^\dagger U_A$. More generally a local trace-preserving clock channel cannot change ρ_P , by the defining property of the partial trace. Dephasing the clock after the pure fixture makes it separable without changing the path signal. Thus a clock's Ramsey decay cannot automatically be multiplied into path visibility. A path-controlled inverse can undo the ideal correlations; implementing it requires controls. Conditional eraser signals must retain their selection probabilities.

4 Numerical hold kernel and a cross-transition consistency relation

Use G-R6's illustrative 1 m height contrast and 300/310 K baths. Let $t_H = 10$ ms be a constant-setting hold, in reference proper-time units, with $\lambda_B = 1$, $\lambda_A = 1 + \gamma$ and $\gamma = g_* \Delta h / c^2$. This is a prescribed hold contribution, not a complete closed interferometer. To avoid subtracting nearly equal optical frequencies, evaluate the algebraically equivalent expression

$$\Delta\nu_i = \nu_i^0[\gamma + (b_{iA} - b_{iB}) + \gamma b_{iA}], \quad \delta_i = 2\pi\Delta\nu_i t_H. \quad (17)$$

The γb_{iA} product is retained as ordinary multiplication; the weak-field background itself is not an all-orders metric.

Hold contrast	Clock	$\Delta\nu$ (Hz)	δ (rad)	$1 - \mathcal{V}$
Gravity only	E2	0.07511	0.004719	$2.784 \cdot 10^{-6}$
	E3	0.07006	0.004402	$2.422 \cdot 10^{-6}$
Thermal only	E2	-0.05051	-0.003173	$1.259 \cdot 10^{-6}$
	E3	-0.006500	-0.0004084	$2.085 \cdot 10^{-8}$

These are ideal two-level-preparation predictions, not observations or achieved precision. The executable also includes identical paths, both controls together, and a genuine shared-ground qutrit. Frequency and phase retain the effect's sign; visibility alone is even in δ and cannot determine that sign.

The same phase data must obey a thermal-rejection identity. Within the retained static response, define

$$x_i = \frac{\delta_i}{2\pi\nu_i^0}, \quad J = \int [\lambda_A u_A - \lambda_B u_B] dt. \quad (18)$$

Equation (3) gives, without a new slowdown function,

$$x_i = \Delta\tau + \beta_i J, \quad \Delta\tau = \int (\lambda_A - \lambda_B) dt. \quad (19)$$

The same G-R5 weights $w_2 = \beta_3/(\beta_3 - \beta_2)$ and $w_3 = -\beta_2/(\beta_3 - \beta_2)$ therefore give

$$w_2 x_2 + w_3 x_3 = \Delta\tau, \quad x_3 - x_2 = (\beta_3 - \beta_2) J. \quad (20)$$

This is a combination of calibrated, unwrapped relative phases, not visibilities. It needs matched trajectories and pulse conventions in the E2/E3 preparations, independent radiometry, and propagated polarizability uncertainty. Two channels cannot independently fit two unknowns and simultaneously validate the law. At matched 300 K and the illustrative hold, the reconstructed $\Delta\tau$ is $1.0911 \cdot 10^{-18}$ s.

The frequency, phase and visibility predictions are distinct observations constrained by one evolution operator, not independent fit functions. In particular a thermally produced gap shift can also produce path-clock correlation. Its mere observation would not uniquely identify gravitational proper time or quantum gravity.

5 What remains physical input, and the next useful test

A hold is not automatically a closed interferometer. The full paths, guiding potentials, launch/recombination dynamics, initialization surfaces and laser phases must be specified. State-dependent guiding and final packet mismatch change the simple two-branch reduction. A standard freely falling light-pulse geometry can cancel the putative uniform-field redshift sensitivity; one cannot substitute $g\Delta h t/c^2$ without checking the whole sequence. Adjustable clock initialization and differential phase protocols offer established alternatives [2]; their neutral-atom pulse design is not assumed to work unchanged for this charged-ion E2/E3 system. The prior G-R6 comparison of two sites is not preparation of one ion in a coherent spatial superposition.

The conservative shift does not determine an open-system visibility. For full branch unitaries W_A, W_B on clock and environment, the cross amplitude is

$$K = \text{tr}_{C,E}(W_A \rho_{CE} W_B^\dagger). \quad (21)$$

Only with initially factorized states and branch factorizations $W_r = U_r \otimes V_r$ does this reduce to κ times a separate environment overlap. Photon which-path records, recoil, thermal correlations and emission generally require the fuller model. Real static polarizabilities constrain a conservative dispersive shift; they do not specify the necessary dissipative correlation functions. No fitted dephasing factor or decay rate is inserted here.

Do not extend the E2 idealization to seconds. The gravity-only first ideal zeros at 1 m would require $t_{\pi,i} = [2\nu_i^0 \gamma(1+b_i)]^{-1}$, approximately 6.66 s for E2 and 7.14 s for E3. Yet the E2 upper state has a tens-of-milliseconds lifetime: a same-isotope study reports 61.8 ms with substantial systematic variation [3]. A newer $^{174}\text{Yb}^+$ study [4] concerns another isotope. We do not pool these values or assign a precise $^{171}\text{Yb}^+$ environmental rate. Even 10 ms is only a conservative hold illustration, not a decay-free experimental claim. E3’s long-lived level would not by itself establish coherent spatial splitting, guide closure or laser coherence.

Bounded conclusion and priority. The unified conservative kernel, pure/mixed counterexample and joint witness are complete under their stated assumptions, with executable density-matrix checks. The missing physical controls are recorded in `CLOCK_PATH_CARD.json`. Next specify a closed path and calibrated state-resolved readout, prioritizing the linear differential phase. Use coherent/dephased preparations and joint measurements to substantiate entanglement. The small-phase visibility loss is quadratic, so it is a consistency check rather than a uniquely diagnostic discovery channel. No data, lab contact, quantum-gravity detection, Ed law or Route-F promotion is supplied by this mathematical extension.

References

- [1] M. Zych, F. Costa, I. Pikovski and C. Brukner, Nature Communications 2, 505 (2011).
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- [3] M. Schacht and M. Schauer, arXiv:1310.2530 (2013), same-isotope lifetime study.
- [4] H. Shao et al., Physical Review Research 5, 023193 (2023).